Introduction to Internet of Things

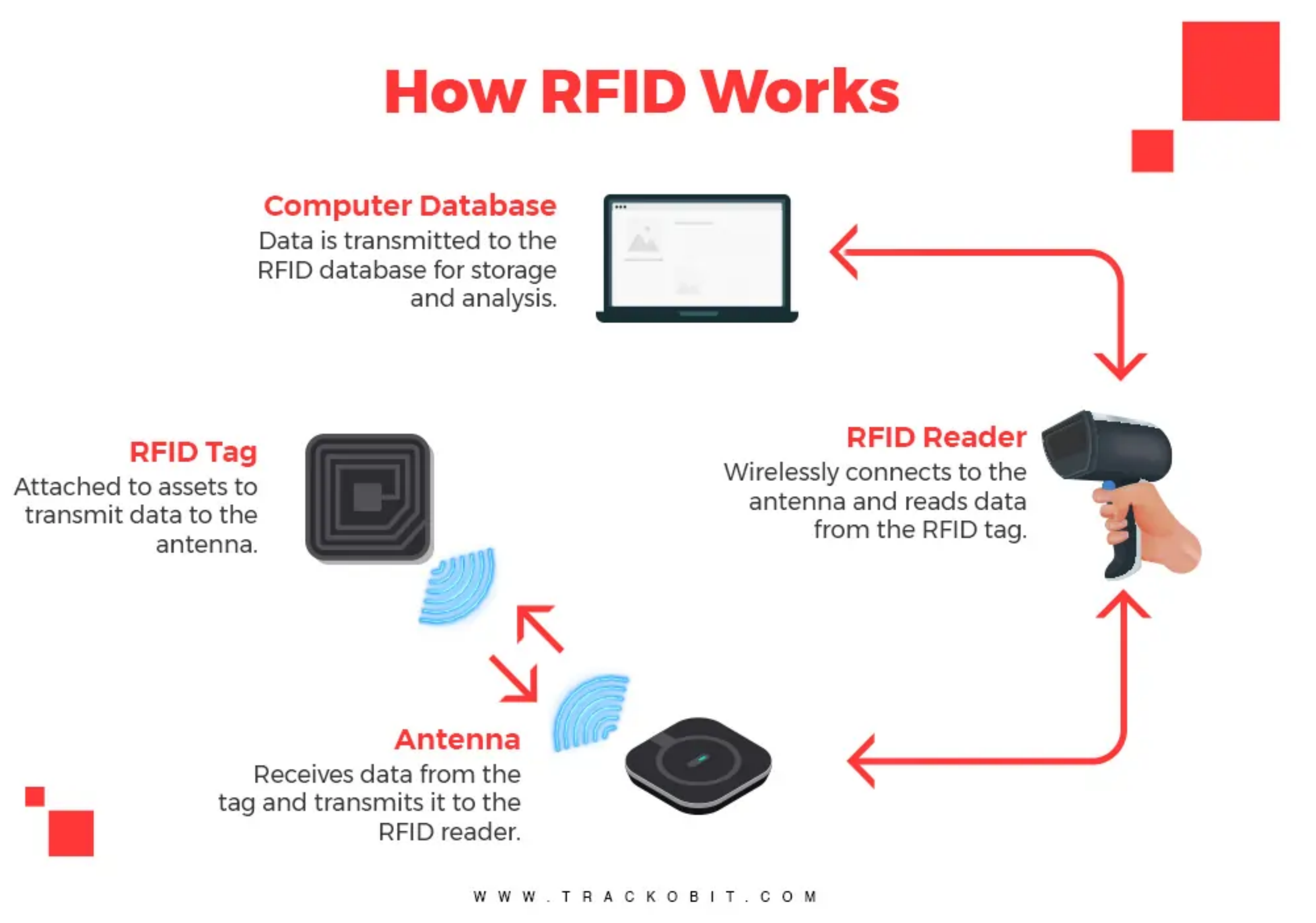
Lecture 20&21

RFID (Radio Frequency identification)

- It is a wireless technology used to identify and track objects using radio waves.
- RFID tags are attached to objects and can be scanned by RFID readers to retrieve information about the object’s location, status, or history.
- RFID systems consist of three main components: tags, readers, and antennas.
- RFID tags are small, wireless devices that can be attached to or embedded in objects.
- RFID readers use radio waves to communicate with the tags and gather data.
- Antennas are used to transmit and receive radio signals between the tags and readers.

Importance of RFID for IoT

- Tracking and Monitoring
- Asset Management
- Automation
- Security
- Analytics



Components

RFID tags:

- RFID tags typically consist of an antenna and a microchip, which contains unique identifying information that can be read by RFID readers.

Passive RFID tags:

- These tags do not have their own power source and rely on the energy from an RFID reader to power the microchip and transmit data.

Active RFID tags:

- These tags have their own power source and can transmit data over longer distances than can passive tags.

Components

RFID Reader:

- An RFID reader is a device that uses radio waves to communicate with RFID tags and collect data.
- RFID readers typically consist of an antenna, a radio frequency (RF) module, and a processor.
- The antenna is used to transmit and receive radio signals, while the RF module converts to digital data that can be processed by the reader's processor.

Fixed RFID:

- These readers are typically installed in a fixed location, such as a warehouse or distribution center.

Handheld RFID readers:

- These are portable devices that can be used to collect data from RFID tags in the field.

Components

RFID Middleware:

- RFID middleware is software that sits between RFID readers and enterprise applications.
- RFID middleware is designed to manage the flow of data between RFID readers and enterprise applications.

RFID-based communication system

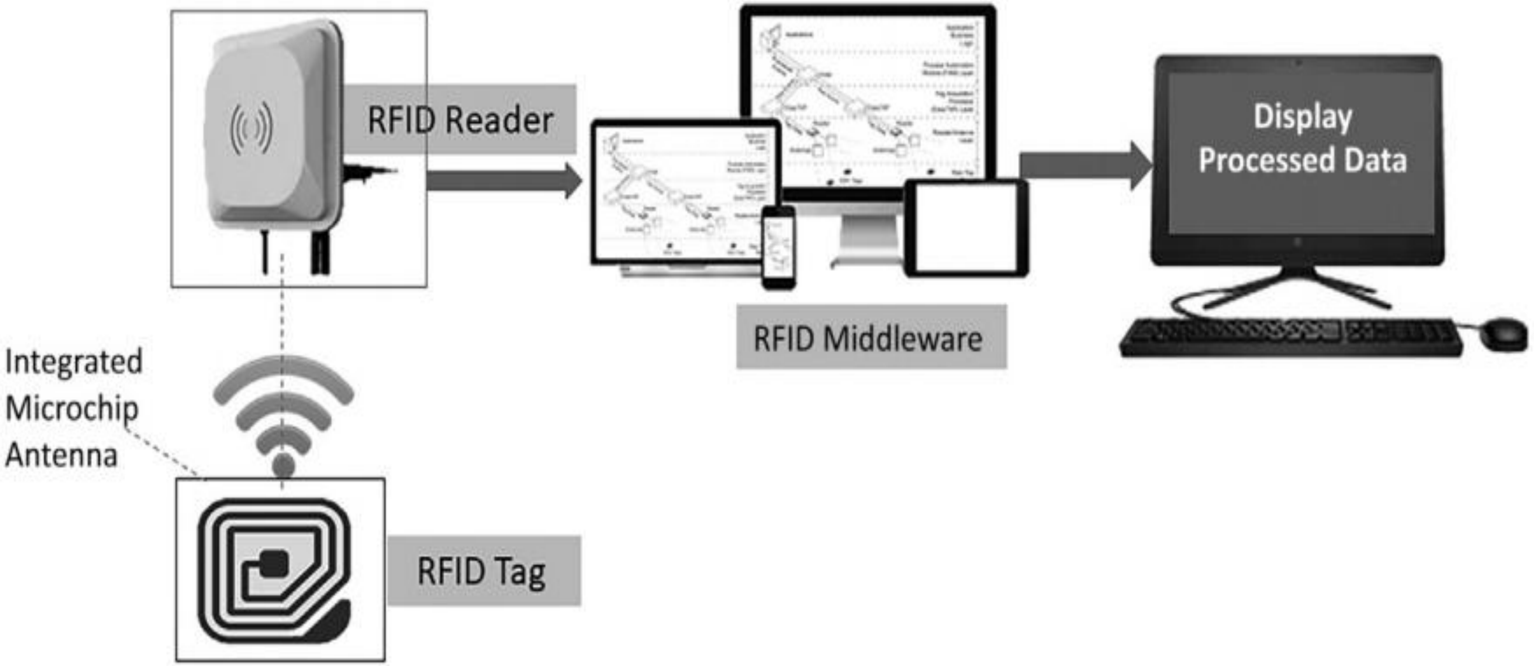
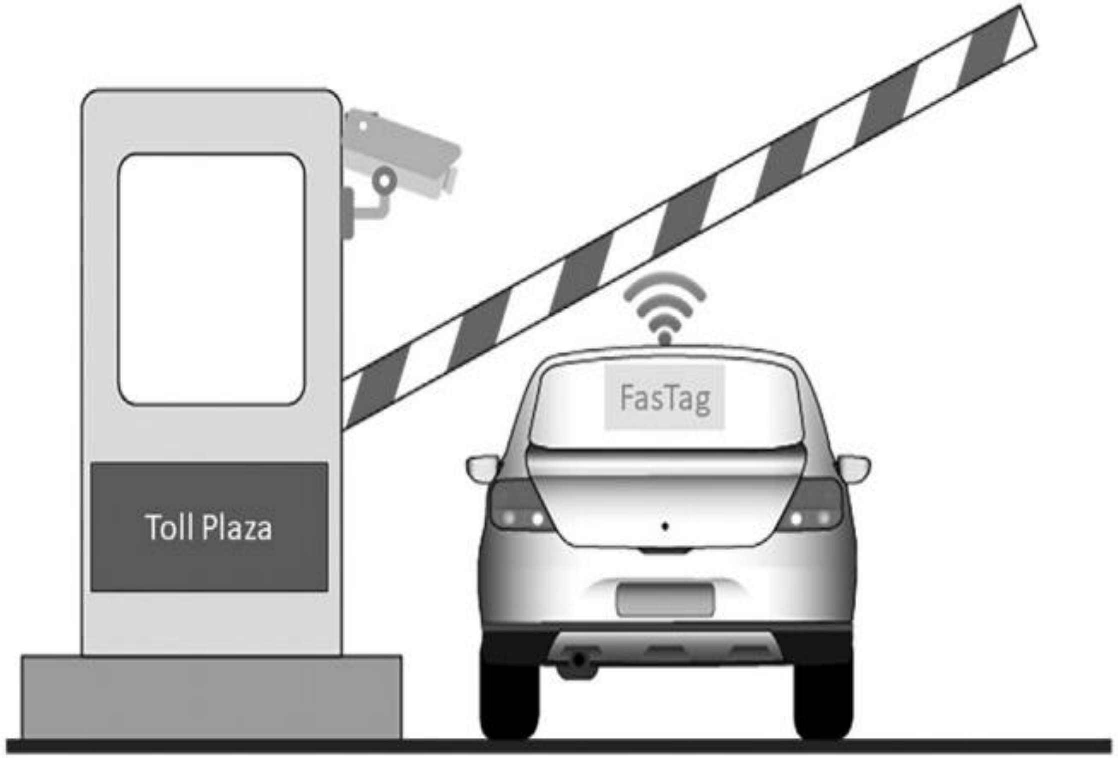


Figure: RFID Tag Communication

Use case



Implementation

a) **RFID Tags:** Each vehicle was equipped with an RFID tag containing a unique identifier. The RFID tags were either affixed to the windshields or integrated into the vehicle registration documents.

b) **RFID Readers:** RFID readers were installed at the entry and exit points of the toll plaza. These readers emitted radio waves and captured information from the RFID tags within their range.

c) **IoT Gateway:** The RFID readers were connected to an IoT gateway, which served as a central communication hub. The IoT gateway collected data from the RFID readers and communicated with the toll plaza's central server.

d) **Real-Time Vehicle Identification:** As a vehicle approached the toll plaza, the RFID reader at the entry point detected the RFID tag and transmitted the vehicle's identification information to the central server. This allowed for real-time vehicle identification and verification.

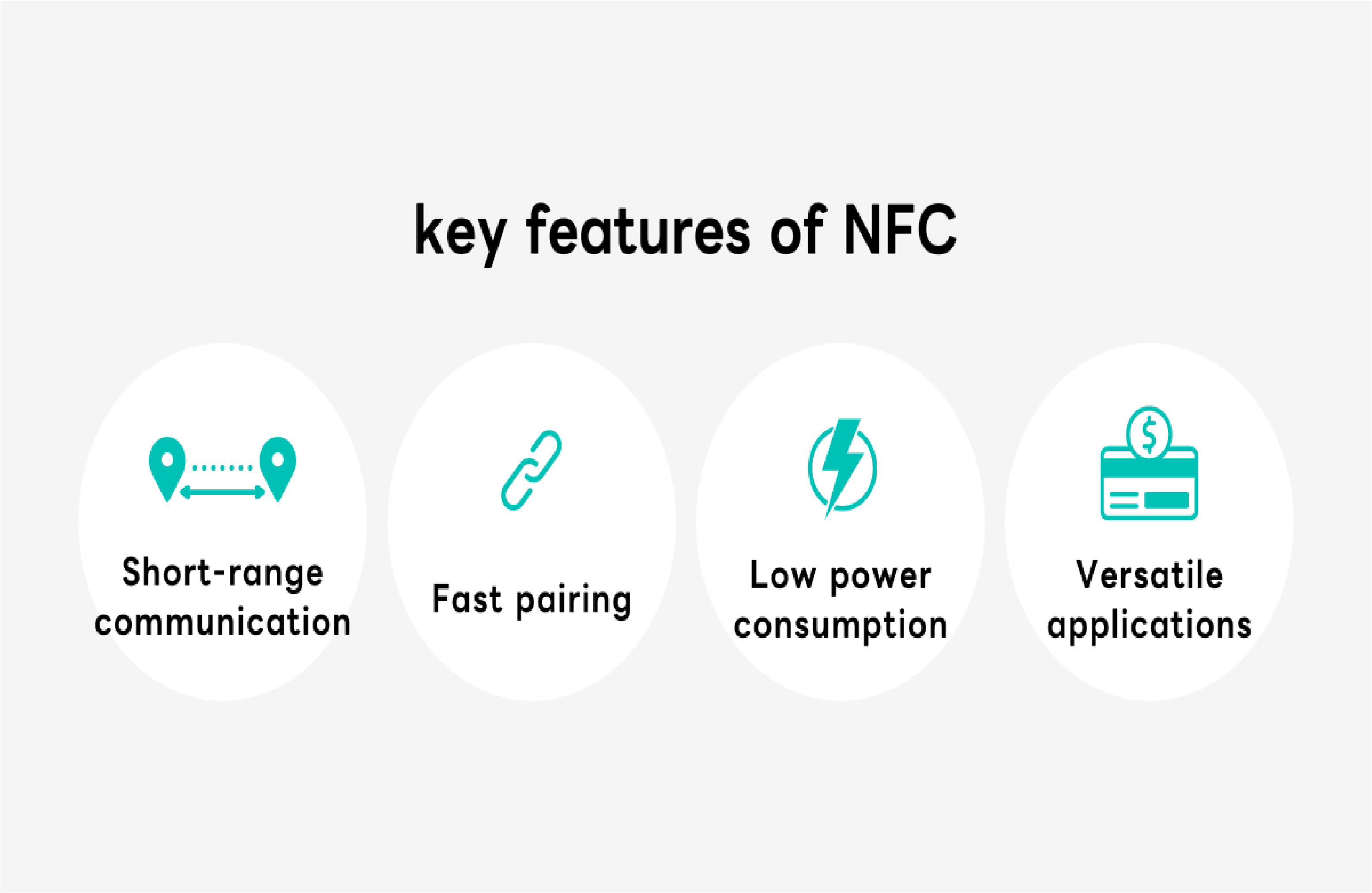
e) **Toll Collection:** Once the vehicle was identified and verified, the toll plaza's system automatically calculated the toll amount based on the vehicle type, distance traveled, and other relevant factors. The driver could then make the payment using cash, a card, or an electronic payment system.
f) **Traffic Monitoring and Management:** The RFID-based IoT communication system also provided real-time traffic data. The RFID readers continuously collected vehicle information, including entry and exit timestamps. This data was analyzed to monitor traffic flow, identify congestion points, and make informed decisions regarding traffic management.
g) **Integration with Other Systems:** The RFID-based IoT system was integrated with other toll plaza systems, such as surveillance cameras, automated barriers, and electronic signage. This integration allowed for seamless coordination and improved operational efficiency.

Benefits

a) **Faster Toll Collection:** The RFID-based system enabled quick and automated vehicle identification, reducing the time taken for toll collection. This resulted in shorter waiting times for drivers and decreased congestion at the toll plaza.
b) **Improved Accuracy:** The automated vehicle identification and toll calculation reduced errors associated with manual collection methods. This improved the accuracy of toll collection and minimized revenue leakage.
c) **Enhanced Traffic Management:** The real-time traffic data collected by the RFID system allowed toll plaza authorities to monitor traffic patterns, identify congestion points, and take proactive measures to manage traffic flow. This helped to reduce congestion and improve overall traffic management.
d) **Convenient and Secure Payments:** The RFID-based IoT system facilitated various payment options, including cash, cards, and electronic payment systems. This provided drivers with convenience while ensuring secure transactions.
e) **Operational Efficiency:** The automated toll collection and real-time data analysis provided by the RFID-based system improved overall operational efficiency at the toll plaza. It reduced the need for manual intervention, minimized errors, and streamlined toll collection processes.

Near Field Communication (NFC)

- It is a wireless communication protocol that is commonly used in IoT applications due to its ability to enable simple, secure, and convenient communication between devices in close proximity.
- Today, NFC is available in almost all mobile, which enables many IoT applications.


key features of NFC
Short-range communication
Fast pairing
Low power consumption
Versatile applications

NFC

- Data is transferred between two (mobile) devices that are NFC enabled or between NFC (mobile) devices and RFID cards, which are close to each other.
- It works on the frequency range of 13.56 MHz.

Passive Mode: In this mode, NFC device does not generate its own radio signal.

It only responds when an external reader powers it using its electromagnetic field.

Example:

Mobile phone used for contactless payment

Metro card or building access card

NFC device modes

Peer-to-Peer Mode: In this mode, two NFC devices can exchange data/ information. As the target device or destination device (listener) uses its power supply, the polling device, i.e., the device which initiates the transmission, requires less power compared to the read/write mode.

Active Mode: In this mode, the NFC device generates its own RF field.

It can read from or write to NFC tags

Example:

- Phone scanning NFC tags in posters
- Access control system reading ID cards
- Inventory or ticket scanning

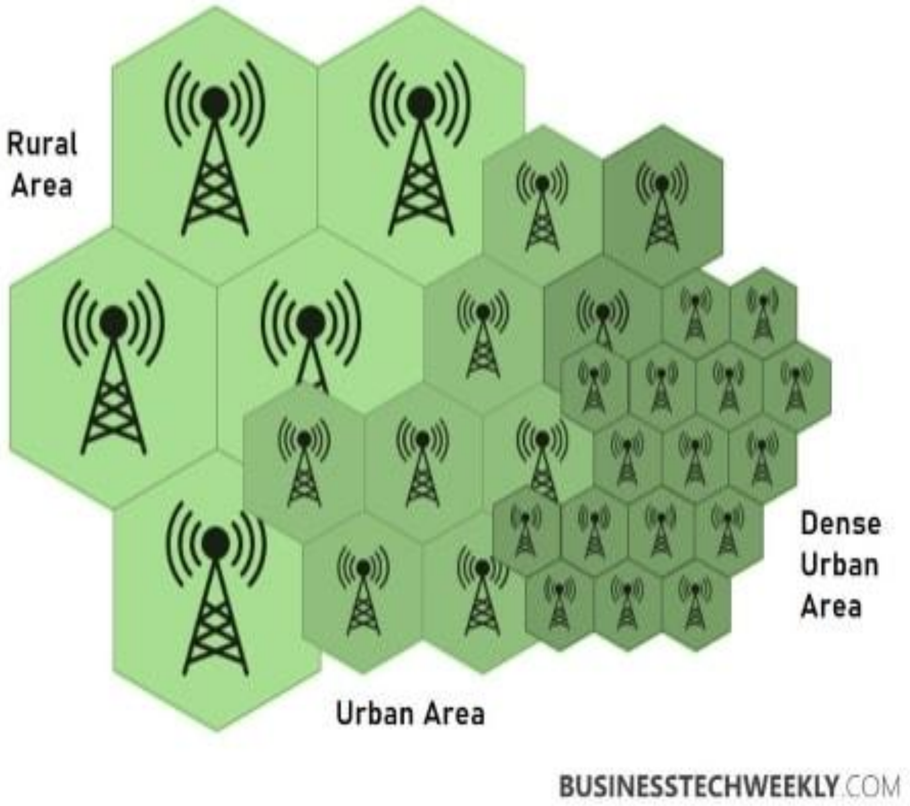
CELLULAR COMMUNICATIONS

In a cellular network, a land area to be supplied with radio service is divided into cells.

The pattern depends on the terrain and reception characteristics.

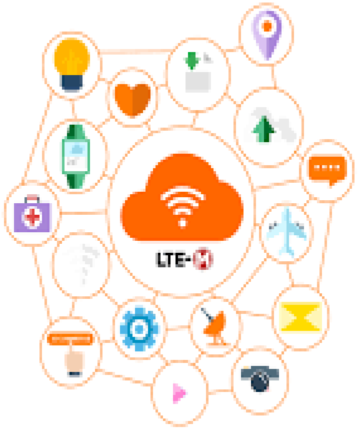
Each of these cells is assigned with multiple frequencies which corresponding radio base stations.

The group of frequencies can be reused in other cells, provided that the same frequencies are not reused in adjacent neighboring cells as that would cause co-channel interference.



Cellular IoT

- Cellular IoT is a class of IoT Connectivity Technology that allows IoT devices to connect to the internet using the same network infrastructure as mobile phones. The growth of the latest connectivity provisions, such as 5G, has allowed cellular IoT to be a considerable choice for **large-scale IoT deployments**.
- It offers very high bandwidth.
- It is used for broadcasting communication.

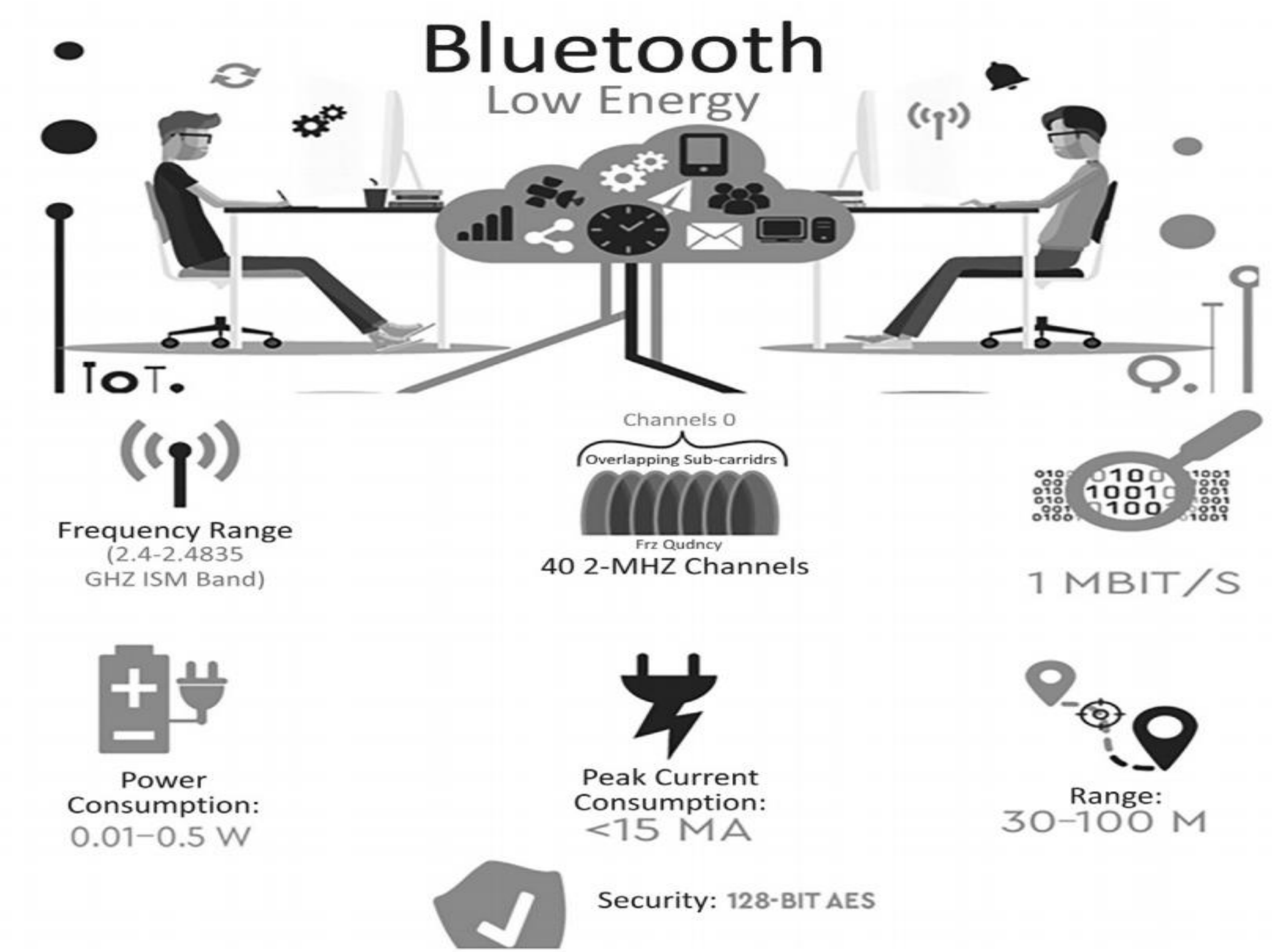


BLE (Bluetooth Low Energy)

- It is a wireless communication protocol that have low power, low cost, and short-range capabilities.
- Its low power consumption, short-range communication, interoperability, security, and compatibility make it a popular choice for IoT developers and organizations.

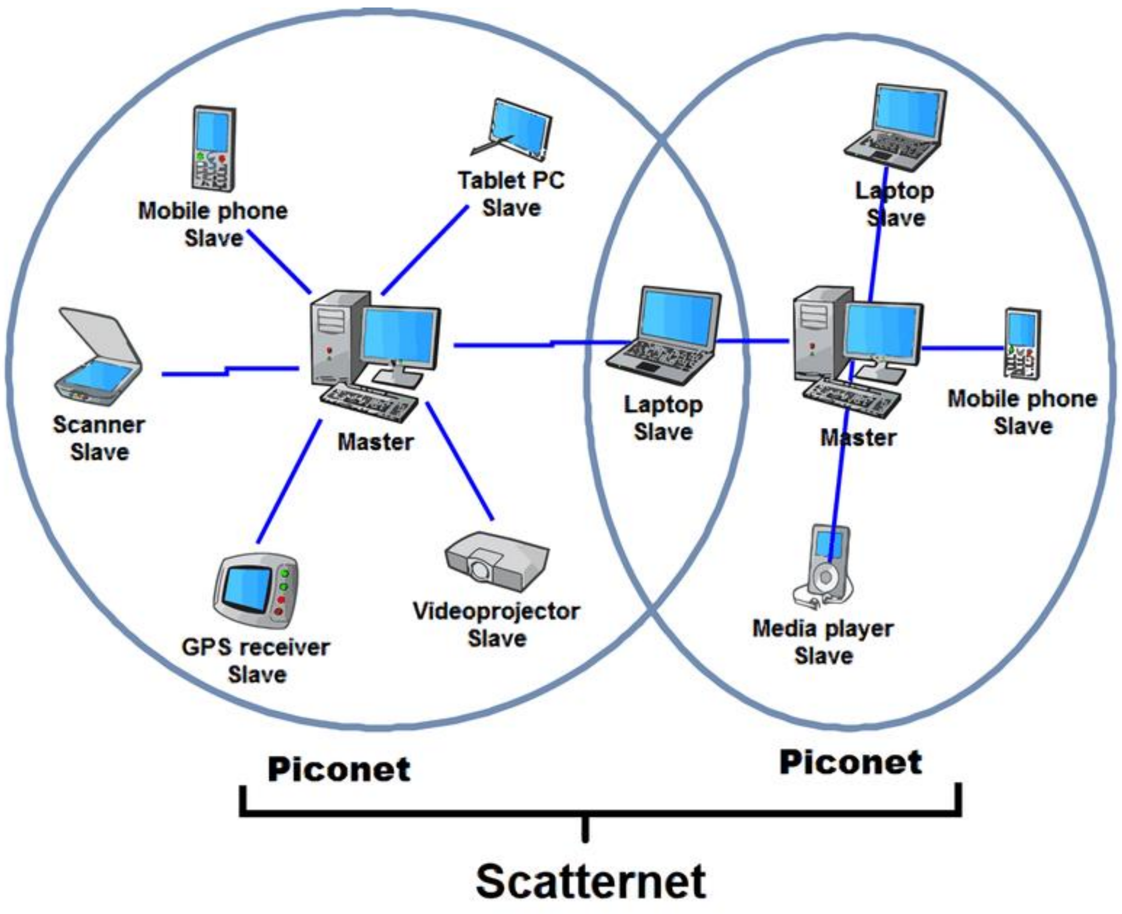
Features of BLE

- Low Power Consumption
- Short-Range Communication
- Interoperability
- Security
- Compatibility


Bluetooth
Low Energy
IoT.
Channels 0
Overlapping Sub-carridrs
Frz Qudncy
Frequency Range
(2.4-2.4835
GHZ ISM Band)
40 2-MHZ Channels
1 MBIT/S
Power
Consumption:
0.01–0.5 W
Peak Current
Consumption:
<15 MA
Range:
30–100 M
Security: 128-BIT AES

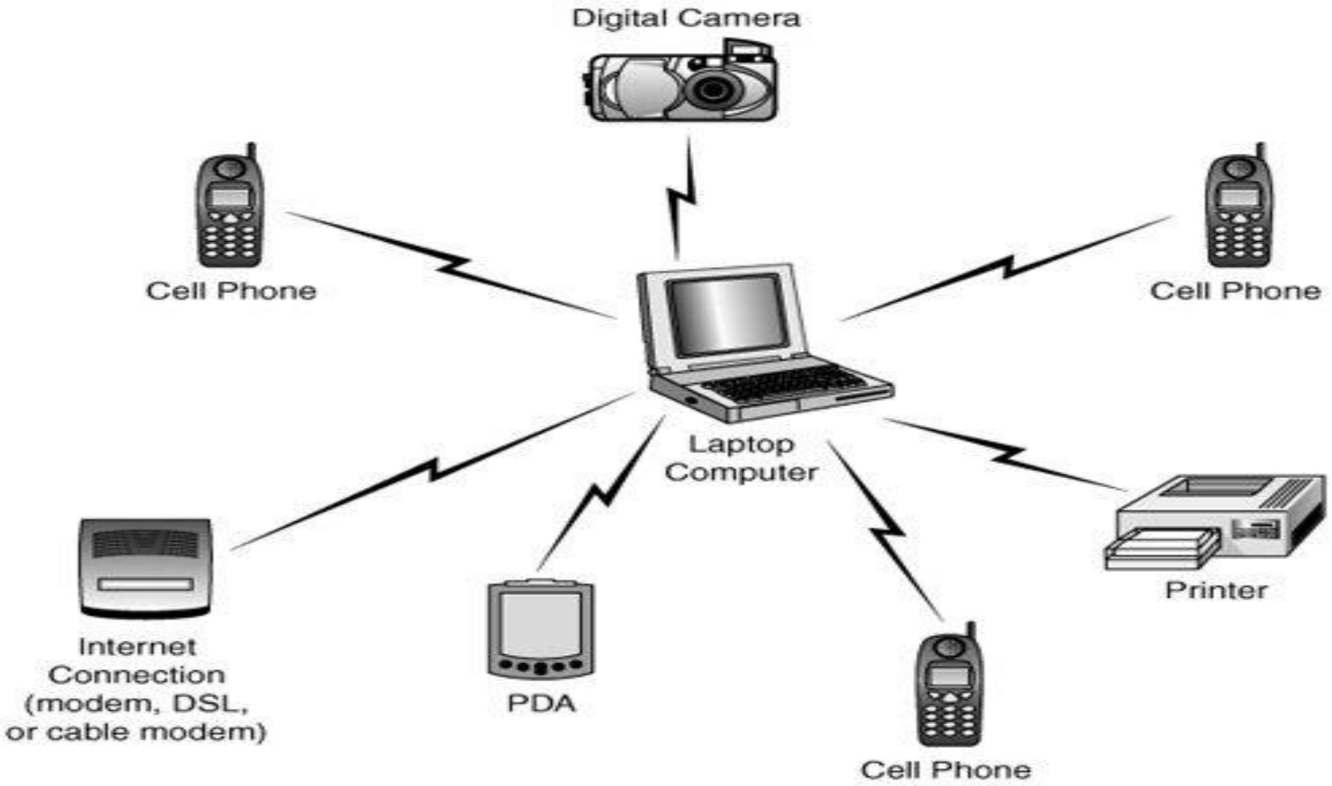
Networks made by Bluetooth

- **Piconet and Scatternet**



BLE Architecture

- The basic BLE netwok configuration is called piconet, where one device acts as a master and the other as slave nodes.
- In piconet, communication is between a master and a slave only.



BLE Architecture

- The collection of piconets is called the scatternet, allowing communication betwene devices in different piconets.